Salient Characteristics

552 Logistics Support Squadron RFID (Radio Frequency Identification)

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1. Systems Interoperability: The software must include tightly integrated hardware modules that have been developed with specified SDK's and API's that allow the hardware to plug and play out of the box. No custom software development services will be allowed on this project. The asset software must be able to be plugged into and recognized on the network by the asset software and become operational immediately upon plug in and configuration through the asset tracking software interface. The following hardware must be included in this capability.

a. 1x Standalone computer
b. 4x RFID Mobile Devices
c. 2x RFID Readers
d. 1x RFID HUB
e. 2x Barcode Printers

Salient Characteristics: The system should be computer-based, and able to function as a stand-alone system for deployment. mapping permissions that must provide warehouse specific rules based on the overall warehouse and the individual sections. Both metallic and plastic surfaces and be resistant to tearing and delaminating. Barcodes must be unique to either the product type or serial number. Barcodes will be a searchable field in the item's stock record. The system must be able to integrate with the barcodes received from product manufacturers. All transactions must be completed by scanning these barcodes. (Scanners should have the ability to operate wirelessly). Issuing and receiving items using barcodes allows personnel to instantly and accurately identify and access the associated stock records. The system must be portable for field operations. The PC must be able to withstand military operations and be military spec compliant. The system must also be able to perform the same barcode functionality with the use of passive RFID tags. These tags must be less than 10x510mm they must be affixed to the assets with an adhesive that can withstand cleaning solvents, corrosive chemicals, extreme cold, extreme heat, submersion in water, and extended deployment times. The RFID system must have the option of providing an RFID infrastructure of fixed readers, antennas, and handheld readers. The following tasks must be completed.

1. Alert system users when items pass a window or door without being on an active AF 1297
2. Check in equipment using only the antenna structure at issuing points
3. Perform inventory operations utilizing RFID antennas and handheld readers, portals spans and arrays

2. The software must include the following capabilities:

A. **Users**: The system must have integrated user capabilities. When creating a system user every task below must be a permission-based capability. The capability should exist where different users would have the ability to create, approve, and fulfill a request for issue. Making system changes should be permission granted by the system administrator. The ability to adjust user permissions is vital in maintaining oversight and accuracy within the systems processes. This area must allow the system administrator to control user access, accounts and permissions.

B. **Personnel**: A dedicated page must show all the personnel who conduct transactions with or through the unit. The receiving personnel should be created into the system. All transactions must be tracked. Personnel should be able to be assigned to their specific squadron or wing within the system.

C. **Receiving Items**: When receiving items into the facility, the system must adhere to all existing Air Force standards and regulations. Upon the completion of every turn in a receipt should be generated to reflect the transaction, this report should be able to be printed and given to the person who returned the item(s). When returning items, you should have the ability to scan barcodes and RFID tags. Every item that is returned should be marked as either serviceable or unserviceable

D. **Hardware**: All hardware should meet compliance as set forth by DOD mandates.

E. **Mobile Operations**: Conducting tasks on a mobile handheld scanner is required. The device must be windows or android based and capable of loading inventory information from the main database. The mobile device must be able to load all pertinent information relating to a stock item to include item name, current quantity on hand values, NSN, serial number, barcode, RFID, GPS IMEI number, and storage location. An inventory must be able to be conducted using the handheld device in an offline mode. Offline mode means the main database information will be exported to the mobile device, the device can inventory all items without being connected, the inventory can then be uploaded into the application and all inventory data will be updated to reflect current levels. The device must be capable of conduct RFID inventories in both offline and online mode.

F. **User Manuals/ Tutorials**: The system must have imbedded training elements. Training videos are preferred. A user manual or tutorial must be present to guide airmen through all processes within the system. Each system task should be complete with pictures and a step-by-step guide to completing. If a system update should be required a manual depicting all software changes is required. These update manuals must follow the same guidelines set forth in the tutorial. Ideally videos would be able to be accessed from each page depicting a page overview as well as how to maneuver through the specific task. All videos, tutorials, and manuals must be viewable when not connected to the internet.

G. **Issuing Items**: Issuing items must be conducted on an independent page. When issuing items, the system must adhere to the standards IAW Air Force instruction. Upon completion of any transaction an electronic Air Force approved hand receipt should be produced. When issuing items to a person the following information must be gathered; personnel name, assigned unit, product(s) received, national stock number (NSN), quantity issued, unit of issue, serial number if applicable, and the transaction date. A digital signature should be captured when the receiving person inputs their CAC and pin. The system user who issued the item(s) to the personnel should also be logged. All the aforesaid requirements must be generated electronically to ensure mistakes are eliminated and records can be maintained. During the issuing process, barcodes and RFID tags should be able to be scanned, assigning the item to the receiving personnel. The software will use serial numbers as the unique identifier to compare barcode, RFID, and GPS devices too. All keys must be based off the serial number. All electronic tagging processes are implemented for speed only. System users need the ability to sign out groups of consumable/ expendable items using a single barcode. A barcode will need to be used when issuing a UTC or increment. A picklist will need to be generated to show the issuing personnel exactly where all of the requested items are located. Lastly, the system should improve the amount of time it takes to issue any number of items to a receiving authority. Although speed is an important aspect, the most vital element is the accuracy provided. Based and capable of loading inventory information from the main database. The mobile device must be able to load all pertinent information relating to a stock item to include item name, current quantity on hand values, NSN, serial number, barcode, RFID, GPS IMEI number, and storage location. An inventory must be able to be conducted using the handheld device in an offline mode. Offline mode means the main database information will be exported to the mobile device, the device can inventory all items without being connected, the inventory can then be uploaded into the application, and all inventory data will be updated to reflect current levels. The device must be capable of conducting RFID inventories in both offline and online mode.

3. The System must provide us with our requirement to have the following tagging options that operate in the same software application.

a. 1D Barcodes
b. 2D Barcodes
c. IUID DOD Barcodes Type 1 and 2
   d. Passive RFID tags (may be added in the future and software must have inherent ability to add these tags upon request)
   e. Active RFID Tags (may be added in the future and software must have inherent ability to add these tags upon request)

4. Tool finder. The system must be able to assist in finding a lost tool using a handheld RFID/RTLS reader. The reader must give a verbal alert when the missing tool is located. System includes this capability.

5. The system must provide RFID inventory for our tool cribs, that system must be able to allow to conduct a tool crib RFID inventory of 5000 items in less than 5 minutes. System has demonstrated that capability.

6. The system must allow for mobile inventory operations while remote using RFID. The system must provide all pertinent tool data and expected locations and allow for multiple types on inventory to be conducted and saved for later import. This inventory RFID must inform the user when items are in the wrong location or when the expected qty is not correct. This is provided by System.

7. The system must provide on the client machines and handheld RFID scanner when scanned items have exceeded shelf life, end of life, scheduled maintenance date, scheduled inventory date, date of manufacture, and usage status. This is a Core feature of System.

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9. The listed tagging capabilities meet our requirement to be able to put any or all of the tags on selected equipment which connect back to the System Database and Unique Item ID.

10. The SYSTEM meets our requirements to have all system transactions for customers and administrators to perform transactions.

11. The SYSTEM meets our need to have all tag options be integrated into one mobile hand-held computer and not require multiple hand-held computers, the hand-held computer must be an android database which our marker research verifies is in use with System.

12. We require that the system include the business rules and work flow for Air Force Research and Development operations and that these specified activities be tracked live in the system. Specifically, we must be able to have the system alert us when specified R&D and T&E tools enter, remain and leave a specified test area. The system must inform us through the tag items that have entered a test location, the time the item spent at each test location and the total time spent in and out of the test location. In addition, we require that the system keep track of the number of times a device has spent in a specified location over a selected period of time. The system must also identify what employee accompanied the items when transitioning across specified choke points and thresholds using RFID/RTLS/GPS tags. System provides this capability.

6.We require that our tool kits be able to be inventoried with and handheld RFID scanner without having to open the tool boxes. The inventory must be on a mobile handheld RFID computer with instantaneous results. This data must sync back to the main database. This inventory must be able to be repeated each time the technician
moves from one repair location to another. This capability exists in the System system.

13. The system must provide tool cart live monitoring while the tool carts are stored in a secure location. The tool cart must be able to be inventoried live while in a locked container from a remote location and not require an onsite person to conduct the inventory. This inventory must be completed in less than 2 minutes for each tool cart. System has demonstrated this capability.